

Kelly Y. Huang

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Kalsi Assistant Professor, Mechanical and Aerospace Engineering
University of Houston

Education

Jul 2021	Ph.D.	Mechanical and Aerospace Engineering	Princeton University
Jan 2018	M.A.	Mechanical and Aerospace Engineering	Princeton University
Dec 2015	B.S.	Mechanical Engineering	Cornell University

Research Interests

Environmental Fluid Mechanics ■ Turbulence ■ Surface-Atmosphere Interactions ■
Boundary-Layer Meteorology ■ Experiments ■ Sensing Techniques

Research Experience

University of Notre Dame

2021 — 2023 **Postdoctoral Researcher** supervised by Prof. Harindra J. S. Fernando
■ fog and turbulence interactions in the marine atmosphere

Princeton University

2016 — 2021 **Graduate Research Assistant** advised by Prof. Marcus Hultmark
■ nano-scale measurements in the atmospheric surface layer
■ active grid for studying mosquito tracking behavior

Cornell University

2015 — 2016 **Undergraduate Research Assistant** advised by Prof. Charles Williamson
■ innovative blade designs for urban vertical-axis wind turbines

National Renewable Energy Laboratory

Summer 2015 **Science Undergraduate Laboratory Intern** advised by Dr. Katherine Dykes
■ optimization of spar supporting structure in offshore wind turbines

Fellowships

2017	National Defense Science and Engineering Graduate Fellowship (~ \$153k) United States Department of Defense
2016	Francis Robbins Upton Fellowship in Engineering (~ \$105k) School of Engineering and Applied Science, Princeton University

Honors and Awards

2025	Teaching Excellence Award Cullen College of Engineering, University of Houston
2024	Faculty-Applied Clean Energy Science (FACES) Program Participant National Renewable Energy Laboratory
2020	Excellence in Teaching Award Engineering Council, Princeton University
2019	The Luigi Crocco Award for Teaching Excellence Mechanical and Aerospace Engineering, Princeton University

Service

2024 – present	Associate Editor; ARC Geophysical Research
2026 – present	Reviewer; Geophysical Research Letters
2024 – present	Reviewer; Physics of Fluids
2021 – present	Reviewer; Experiments in Fluids
2024, 2025	Conference organizer; American Physical Society Division of Fluid Dynamics (APS DFD) Annual Meeting

Outreach

2025	Panelist, Fluids Education Lunch – <i>APS DFD</i>
2024	Mentor, Student Networking Lunch – <i>APS DFD</i>
2021	Trained in Inclusive Leadership Certificate program – <i>Princeton University</i>
2021	Panelist, women-in-STEM discussions – <i>Rutgers University</i>
2016 – 2021	Led K–12 STEM outreach and developed hands-on demos to counter engineering stereotypes – <i>Harlem Preparatory School in New York City, French American Elementary School of Princeton.</i>
2016	Completed "Inspiring Young Engineers – Outreach" course – <i>Princeton University</i>

Professional Memberships

2016 – present	American Physical Society (APS)
2018 – present	American Geophysical Union (AGU)

Publications

*supervised students are underlined; *denotes undergraduate student*

- Peer-Reviewed F. Pesenti, H. Motalebi, R. Niraula, G. G. Katul, **K. Y. Huang**, “Directional Dynamics of Fog: Irreversibility and Causal Coupling with Turbulence,” *Geophysical Research Letters* (2026).
- K. Y. Huang**, R. Niraula, G. G. Katul, “Scaling laws for the longitudinal heat turbulent flux in the atmospheric surface layer,” *Journal of Geophysical Research: Atmospheres* (2026).
- H. J. S. Fernando, ... , **K. Y. Huang**, ..., “Fatima-GB: Searching Clarity within Marine Fog,” *Bulletin of the American Meteorological Society* (2025).
- T. J. Hintz, **K. Y. Huang**, S. W. Hoch, J. Ruiz-Plancarte, and H. J. S. Fernando, “A mechanism for coastal fog genesis at evening transition,” *Quarterly Journal of the Royal Meteorological Society* (2024).
- K. Y. Huang**, G. G. Katul, T. J. Hintz, J. Ruiz-Plancarte, and H. J. S. Fernando, “Fog intermittency and critical behavior”, *Atmosphere* (2023).
- H. J. S. Fernando, S. Wang, **K. Y. Huang**, and E. Creegan, “Fog-laden density staircases in marine atmospheric boundary layer”, *Environmental Fluid Mechanics* (2023).
- K. Y. Huang**, M. K. Fu, C. P. Byers, A. D. Bragg, and G. G. Katul, “Logarithmic scaling of higher-order temperature moments in the atmospheric surface layer”, *International Journal of Heat and Fluid Flow* (2023).
- K. Y. Huang** and G. G. Katul, “Profiles of high-order moments of longitudinal velocity explained by the random sweeping decorrelation hypothesis”, *Physical Review Fluids* (2022).
- K. Y. Huang**, C. E. Brunner, M. K. Fu, K. Kokmanian, T. Morrison, A. O. Perelet, M. Calaf, E. Pardyjak, and M. Hultmark, “Investigation of the Atmospheric Surface Layer Using a Novel High-resolution Sensor Array”, *Experiments in Fluids* (2021).
- K. Y. Huang**, G. G. Katul, and M. Hultmark, “Velocity and temperature dissimilarity in the surface layer uncovered by the telegraph approximation”, *Boundary-Layer Meteorology* (2021).
- Conference Proceedings N. Durand*, H. Motalebi, C. Byers, **K. Y. Huang**, “Frequency-dependent spectral structures in active grid turbulence”, *14th Int. Symp. on Turbulence and Shear Flow Phenomena, Heidelberg, Germany* (2026).
- I. Decan*, R. Niraula, C. Byers, **K. Y. Huang**, “Replicating canopy turbulence with a novel forest analog”, *14th Int. Symp. on Turbulence and Shear Flow Phenomena, Heidelberg, Germany* (2026).

K. Y. Huang, M. K. Fu, C. P. Byers, and G. G. Katul, “Logarithmic scaling of higher-order temperature moments in the atmospheric surface layer”, *12th Int. Symp. on Turbulence and Shear Flow Phenomena, Osaka, Japan* (2022).

Conference Presentations

*supervised students are underlined; *denotes undergraduate student*

Poster	<u>H. Motalebi</u> , <u>F. Pesenti</u> , K. Y. Huang , “Causal Relationships between Fog and Turbulence”, <i>American Meteorological Society Annual Meeting</i> (2026).
Poster	<u>R. Niraula</u> , <u>I. Decan</u> *, C. Byers, K. Y. Huang , “Replicating Canopy Turbulence with a Novel Forest Analog”, <i>American Meteorological Society Annual Meeting</i> (2026).
Poster	<u>R. Niraula</u> , K. Y. Huang , G. G. Katul, “Characteristics of the longitudinal heat flux co-spectrum in stratified atmospheric flows”, <i>American Geophysical Union Annual Meeting</i> (2025).
Talk	<u>R. Niraula</u> , K. Y. Huang , G. G. Katul, “The role of thermal stratification on the longitudinal heat flux co-spectrum”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Interact	<u>H. Motalebi</u> , H. J. S. Fernando, I. Gultepe, K. Y. Huang , “Hot-wire anemometry for simultaneous droplet and turbulent measurements”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Poster	<u>H. Joshi</u> , <u>S. Ortiz</u> *, S. Mohapatra, W. Shih, K. Y. Huang , “Development of a highly sensitive, fast response surface plasmon-based humidity sensor”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Poster	<u>S. Bhamare</u> *, <u>S. E. Ancin</u> *, K. Y. Huang , M. Fu, C. Byers, “Development of a cost-effective constant temperature hot wire anemometer”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Poster	<u>N. Durand</u> *, <u>H. Motalebi</u> , C. Byers, K. Y. Huang , “Characterizing flow with an active grid”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Poster	<u>I. Decan</u> *, <u>R. Niraula</u> , K. Y. Huang , C. Byers, “Flow profiles behind a novel model forest canopy”, <i>Annual Meeting of the Division of Fluid Dynamics</i> (2025).
Poster	K. Y. Huang , <u>F. Pesenti</u> “Fog Causality, Reversibility, and Formation Mechanisms”, <i>American Geophysical Union Fall Meeting</i> (2024).
Poster	J. Ruiz-Plancarte, R. Yamaguchi, S. W. Hoch, K. Y. Huang , H. J. S. Fernando, Q. Wang, “Surface layer properties and fog processes of Sable Island during FATIMA”, <i>American Geophysical Union Fall Meeting</i> (2024).
Talk	<u>F. Pesenti</u> , K. Y. Huang , “Fog Causality, Reversibility, and Formation Mechanisms”, <i>APS Division of Fluid Dynamics Meeting</i> (2024).
Talk	C. Byers, K. Y. Huang , M. Fu, “Corrections for higher-order spectral estimates of Gaussian processes”, <i>APS Division of Fluid Dynamics Meeting</i> (2024).

Poster U. Balasuriya*, **K. Y. Huang** “The development of a low-cost system for measuring pressure fluctuations and transport over complex terrain”, *APS Division of Fluid Dynamics Meeting* (2024).

Invited Seminars

2026	The Pennsylvania State University, University Park, USA — Aerospace Engineering
2025	Max Planck Institute for Dynamics and Self-Organization, Göttingen, Germany Laboratoire des Écoulements Géophysiques et Industriels, Grenoble, France
2024	University of Maryland, College Park, USA — Aerospace Engineering University of Houston, Houston, USA — Civil and Environmental Engineering
2023	Duke University, Durham, USA — Civil and Environmental Engineering U.S. Naval Academy, Naval Academy, USA — Mechanical Engineering National Taiwan University, Taiepi, Taiwan — Hydrotech Research Institute National Central University, Taoyuan City, Taiwan — Civil Engineering
2021	Univ. of Notre Dame, Notre Dame, USA — Environmental Fluid Dynamics Seminar University of California, Davis, USA — Environmental Dynamics Lab Seminar
2020	Cooper Union, New York City, USA — Albert Nerken School of Engineering

Teaching

University of Houston

2025 — present	Pathway Professor, appointed to teach foundational engineering courses critical to student retention and success.
Spr 26	■ MECE 6384 – Methods of Applied Mathematics I
Spr 24 & 25	■ MECE 2334 – Thermodynamics
Fall 24	■ MECE 5397/6397 – Introduction to Environmental Fluid Dynamics

Princeton University

2017 — 21	Graduate Coordinator for the McGraw Learning and Tutoring Center Assistant in Instruction
Fall 19	■ MAE 305/MAT 391 – Mathematics in Engineering I
Spr 19	■ MAE 222 – Introduction to Fluid Mechanics
Spr 18	■ MAE 224 – Integrated Engineering Science Laboratory
Fall 17	■ MAE 335 – Fluid Dynamics
	Guest Lecturer
Fall 22	■ MAE 551 – Fluid Dynamics
Spr 22	■ MAE 553 – Turbulence

Cornell University

	Undergraduate Teaching Assistant
Fall 15	■ MAE 3230 – Introduction to Fluid Mechanics
Fall 15	■ MAE 6510 – Advanced Heat Transfer

Spr 15 ■ MAE 2250 – Mechanical Synthesis
Fall 14 ■ ENGRD 2020 – Statics and Mechanics of Solids

University of Notre Dame

Guest Lecturer

Fall 21 & 22 ■ CE/AME 40465/60465 – Mechanics of Environmental Motions
Fall 21 – Spr 23 ■ CE 62400 – Environmental Fluid Dynamics Practicum
Spring 23 ■ CE 60430 – Fundamentals of Turbulence Theory

Student Thesis Supervision

2023 Hintz, Thomas J. — M. S., University of Notre Dame
 A Mechanism for Coastal Fog Genesis at Evening Transition

2024 Pesenti, Filippo — M. S., University of Houston
 Fog Causality, Reversibility, Forming Mechanisms, and Causal Relationships with Turbulence